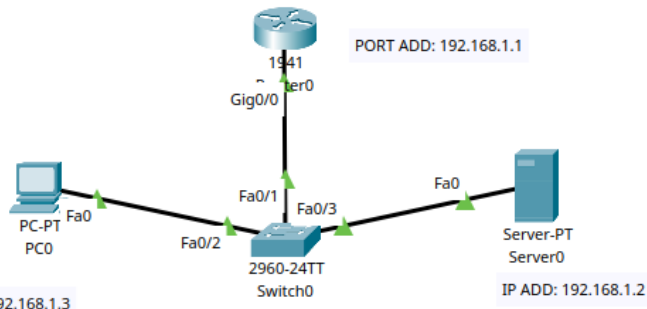




Router0

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started!

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#aaa new-model
Router(config)#radius-server host 192.168.1.2 key cisco
Router(config)#aaa authentication login default group radius local
Router(config)#username backup privilege 15 secret backup123
```

```
C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...Open
```

User Access Verification

```
Username: admin
Password:
% Login invalid
```

User Access Verification

```
Username:
User Access Verification
```

```
Username: admin
Password:
Router>
```

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>telnet 192.168.1.1
```

Server0

Physical Config Services Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

AAA

Service ☒ On ☐ Off Radius Port

Network Configuration

Client Name Client IP

Secret ServerType

	Client Name	Client IP	Server Type	Key	
1	R1	192.168.1.1	Radius	cisco	Add
					Save
					Remove

User Setup

Username Password

	Username	Password	
1	spandan	spandan123	Add
			Save
			Remove

☐ Top

```
Router(config)#ip domain-name lab.local
Router(config)#hostname R1
R1(config)#crypto key generate rsa
The name for the keys will be: R1.lab.local
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#ip ssh version 2
*Mar 1 0:15:25.606: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 0 5
R1(config-line)#transport input ssh
R1(config-line)#login authentication default
R1(config-line)#exit
R1(config)#
```

```
Username: admin
Password:
Router>ssh -l admin 192.168.1.1

Password:
R1>
```